

COST-BENEFIT MEMO

To: Dr. De Freitas, ED Board & Martina Griffith (Clinical IT Lead), Mercer General Hospital

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Re: Model Optimisation & Trade-offs — Recommendation for Phase 3

Verdict

We recommend Random Forest with ESI-1 probability threshold tuning for Phase 3, it raises critical patient recall from 0.500 (baseline) to 0.750, correctly catching 12 of 16 ESI 1 patients in the held-out test set, at a compute and interpretability cost that Mercer General can practically manage within current IT infrastructure and clinical governance constraints.

1. Dataset & Methods Recap

The model was trained on the Yale EMLC ED Triage Dataset: 55,121 adult ED encounters from Yale New Haven Hospital, Connecticut, USA. Seven vital signs, patient age, and 51 chief complaint binary flags (prevalence $\geq 0.5\%$) were used as inputs, 59 features total. Two columns (disposition, previousdispo) were excluded as leakage risks, known only after triage.

Class imbalance: ESI 3 dominates at 49.0% of all visits; ESI 1, the most critical patients requiring immediate resuscitation, accounts for only 77 patients (0.14%) across the full dataset, with 16 in the held-out test set. This extreme imbalance means standard model training ignores ESI 1 entirely. The breakthrough this week was threshold tuning: rather than relying on hard class predictions, we lower the ESI-1 probability bar so the model flags patients it is moderately confident about, rather than requiring majority confidence that 0.14% prevalence makes impossible to achieve.

Population transfer gap: This is a US academic hospital dataset (Yale New Haven; 53.4% White/Caucasian). Yang et al. (2024) demonstrated that direct deployment of a high-income country model to a low/middle-income ED without local adaptation produces poor and highly variable performance. All results here must therefore be treated as a proof-of-concept floor, not a deployment-ready ceiling, until Mercer General's own EHR data is available for local validation.

Train/test split: 80/20 stratified split, random_state=27, fixed throughout. All three complex models (Random Forest, Gradient Boosting, MLP) were evaluated on the identical 11,025-patient held-out test set as Week 6 baselines, re-shuffling the split would make cross-week comparison dishonest.

2. Six-Axis Benchmark Table

Primary metric ★: ESI-1 Recall — of all patients who truly needed immediate resuscitation, what proportion did the model correctly flag? A false alarm (predicting ESI 1 for an ESI 2–3 patient) wastes resources but is correctable by the triage nurse. A missed ESI 1 patient waits in the queue while their condition deteriorates, in a Caribbean ED like Mercer General, operating at reduced staffing (documented night shifts of 4 nurses against a target of 14 per De Freitas et al., 2020), that window may be irreversible.

Note: Gradient Boosting and MLP rows marked * will populate when the notebook is run. Random Forest values confirmed from notebook output.

Model	Accuracy	Macro F1	ESI-1 Recall ★	Train	Infer (ms)	Interpret.
Dummy (stratified)	0.372	0.200	0.000	<0.1s	<0.001	N/A
Logistic Regression (Wk 6)	0.457	0.345	0.500	~7.7s	0.0011	High
Decision Tree (Wk 6)	0.375	0.261	0.125	0.34s	0.0005	High
Random Forest + Threshold	0.500	0.357	0.750	~35s	0.0747	Medium
Gradient Boosting + Threshold	0.627	0.366	0.188	~4min	0.0185	Low
MLP + Threshold	0.444	0.311	0.500	~2min	0.0025	Low

Threshold tuning note: The ESI-1 decision threshold for each model is derived from the training set ESI-1 probability distribution only. No test data was used to set thresholds, preserving evaluation integrity.

3. Cost-Benefit Comparison

Dimension	Logistic Regression	Random Forest + Threshold	GB + Threshold	MLP + Threshold
ESI-1 Recall ★	0.500	0.750	0.188	0.500
Interpretability	High	Medium	Low	Low
Training cost	~8s	~35s	~4 min	~2 min
Inference (ms/pred)	0.001	0.075	~0.1	~0.002
IT governance risk	Low	Medium	Medium-High	High
Audit trail	Easy	Moderate	Hard	Hard
Fairness audit	Straightforward	Per-subgroup importances	Needs SHAP	Needs SHAP

4. Three Arguments FOR Random Forest + Threshold

4.1 ESI-1 Recall improvement is clinically significant

Random Forest + threshold correctly identifies 12 of 16 critically ill patients (0.750 recall), compared to 8 of 16 for Logistic Regression (0.500). In a Caribbean ED seeing approximately 150 patients per day with an ESI 1 prevalence of 0.14%, this translates to roughly one additional critical patient correctly flagged per day. In an under-resourced ED where re-assessment intervals can be long, that gap is the difference between intervention and deterioration.

4.2 Medium interpretability is governable at Mercer General

Random Forest sits at Medium interpretability, below Logistic Regression but above Gradient Boosting and MLP. Feature importances are globally available and can be reviewed by the Clinical AI Lead. Unlike Gradient Boosting and MLP, Random Forest does not require SHAP tooling to provide any accountability trail. This is significant given Mercer General's documented racial fairness concern: Black/African American patients are allocated to ESI 4 at 20.5% versus 12.8% for White/Caucasian patients (Week 5 feasibility memo). A model that can be interrogated at feature level, even without per-patient SHAP, is better positioned for the subgroup equity audit required before deployment.

4.3 Inference cost is manageable at Mercer General's patient volume

Random Forest inference takes approximately 0.075 ms per prediction. At 150 patients per day, that is 11.25 ms total inference time daily, negligible even on Mercer General's existing hardware. Martina Griffith's governance concern about hidden compute costs is legitimate, but at this patient volume, Random Forest's inference cost does not compound into an operational burden. Training is a one-off cost of approximately 35 seconds, paid only on retraining.

5. Three Arguments AGAINST Random Forest + Threshold

5.1 0.750 recall still means 4 in 16 critical patients missed

Even at the improved threshold, 4 ESI 1 patients in the test set were missed. These patients tend to present with near-normal vital signs, early septic shock with compensatory haemodynamics, ambulatory intracranial bleeds, the same presentations a tired nurse might miss. The model cannot be deployed as an autonomous triage agent; it must function as a decision support tool only, with the triage nurse retaining final authority. This is non-negotiable.

5.2 Threshold tuning reduces precision and may increase alert fatigue

Lowering the ESI-1 decision threshold increases recall but also increases false positives, patients flagged as ESI 1 who are actually ESI 2 or 3. In a busy ED already operating at reduced staffing, false alarms erode clinical trust rapidly (Risk 6 in the Week 4 risk register: Alert Fatigue, rated High likelihood, High impact). The threshold must be monitored monthly and adjusted by the Clinical AI Lead as patient volume and demographics shift.

5.3 Population transfer gap limits confidence in these metrics

All benchmark figures are derived from a US academic hospital dataset. Mercer General's patient population, with higher prevalence of vector-borne illness, non-communicable disease burden, and walk-in presentations, may produce different feature importance rankings and different ESI 1 presentations than the Yale training distribution. Yang et al. (2024) showed this gap is real and can produce highly variable performance. The 0.750 recall figure should be treated as an optimistic ceiling until local validation is complete.

6. Risks & Unknowns

EHR data lag (Technical constraint, Week 4): Vitals at Mercer General are recorded on paper first and entered into the EHR 5–30 minutes later. Plug-in 2 (ESI prediction) cannot pull live EHR data at the point of nurse assessment, it must use a manual-entry interface. Random Forest's 0.075 ms inference time is compatible with this; it can run on bedside-entered vitals before the EHR record is complete.

Nurse time pressure (Operational constraint, Week 4): Triage nurses assess 15–30 patients per shift, frequently understaffed. Any alert requiring an extra click or mandatory override step per patient competes directly with bedside care. The model output must default to passive display, visible alongside the nurse's own assessment, not an active step the nurse must clear.

Handover information loss (Structural constraint, Week 4): De Freitas et al. (2020) identified shift handover as the highest information-loss point in the ED flow. An ESI 1 flag raised by the model must persist as a visible, timestamped field in the patient chart, not only as a dashboard alert, to survive handover.

What we do not yet know: (1) How Mercer General's triage nurses will respond to model alerts in real practice, whether they will trust it, ignore it, or integrate it appropriately. (2) Whether the threshold that achieves 0.750 recall on Yale data will hold on Mercer General's patient presentations. Both can only be assessed once the pilot begins.

7. Recommendation

Proceed to Phase 3 with Random Forest + ESI-1 threshold tuning as the recommended model, deployed as a passive decision support tool alongside triage nurse assessment, subject to: (1) monthly ESI-1 recall monitoring with a 3 percentage point drop trigger for mandatory review; (2) subgroup performance audit by race, gender, and arrival mode before go-live; and (3) local validation on Mercer General EHR data before any clinical deployment recommendation is made to the Saint Cedric Ministry of Health.